

El Moustpha Cheikh JIDDOU

Engineering Student at the Higher Polytechnic School

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PROFESSIONAL EXPERIENCE

Data & AI Engineer – Richat Partners

📅 August 2026 – Present

📍 Nouakchott, Mauritania

- Design and implementation of AI and data solutions for strategic consulting missions.
- Development of data pipelines and machine learning models to support decision-making in digital transformation, investment, and strategy projects.
- Collaboration with multidisciplinary teams to translate business needs into technical solutions.
- Contribution to the firm's AI offerings: LLM integration, RAG systems, predictive analytics, and data-driven advisory.

Data Systems & IT Architecture Intern – Banque Nationale de Mauritanie (BNM) 📅 February 2025 – July 2026

📍 Nouakchott, Mauritania

- Designed and implemented a centralized enterprise Data Warehouse backed by robust ELT pipelines, automated flow orchestration, and predictive models.
- Architected and deployed a containerized microservices infrastructure over isolated Docker networks on on-premise production servers.
- Built a secure conversational agent (Click) connected to the WhatsApp REST API, including a custom NLU module for dynamic query routing.
- Implemented a RAG (Retrieval-Augmented Generation) architecture with a PostgreSQL/pgvector database to index and query official banking documentation without data leakage.
- Benchmarked several large language models (GPT-4o, Claude 3.5, Gemini 1.5, local Llama) to optimize the trade-off between performance, infrastructure cost, and strict confidentiality.
- Developed sequential neural network models (LSTM, GRU) to anticipate rapid photovoltaic power drops and plan for critical infrastructure resilience.
- Technical environment:** Python (FastAPI), Data Warehousing, ELT pipelines, Docker, N8N, PostgreSQL (pgvector), PyTorch, React, REST APIs, Linux/Bash.

End-of-Studies Internship – Macroeconomic Analysis and DSGE Modeling – Banque Centrale de Mauritanie (BCM) – Directorate General of Economic Studies and Monetary Stability 📅 February 2026 – May 2026

📍 Nouakchott, Mauritania

- Produced a master's thesis in Econometrics and Applied Statistics on the comparative analysis of monetary regimes in Mauritania using a small open-economy DSGE model.
- Analyzed monetary policy transmission mechanisms, inflation, exchange rate dynamics, and macroeconomic dynamics in the Mauritanian context.
- Structured and exploited macroeconomic and monetary time series from BCM, ANSADE, and international databases for calibration and economic simulations.
- Participated in the activities of the Directorate General of Economic Studies and Monetary Stability: conjunctural analysis, macroeconomic forecasts, monetary statistics, and economic studies.
- Developed economic modeling and simulation tools to compare several monetary policy regimes and assess their impact on macroeconomic stability.
- Produced quantitative analyses, economic charts, and statistical indicators to support results interpretation and decision-making.
- Environment:** Econometrics, DSGE, macroeconomic analysis, monetary statistics, Python, R, Excel, Julia, GAMS, economic modeling, data analysis.

Lecturer – Poverty & Household Living Conditions + Sociological Survey Methods – Institut Supérieur des Métiers de la Statistique (ISS) 📅 February 2026 – June 2026

📍 Nouakchott, Mauritania

- Renewed for a second consecutive year with an additional course module.
- Taught the Poverty and Household Living Conditions module (improved edition).
- Designed and delivered a new module: Sociological Survey Methods (survey methodology, sampling, questionnaire design, data analysis).
- Supervised tutorials and practical sessions with field survey simulations using KoboToolbox, CSPro, and Stata.
- Directed and evaluated final-year projects (PFE) and supervised worker internships.
- Developed course materials, case studies, and evaluation frameworks for both modules.

Lecturer – Poverty and Household Living Conditions – Institut Supérieur des Métiers de la Statistique (ISS) 📅 February 2025 – June 2025

📍 Nouakchott, Mauritania

- First higher education teaching experience at the institution from which I graduated.
- Prepared and delivered lectures on poverty measurement methodologies and household living conditions analysis.
- Designed and supervised tutorials (TD) and practical sessions (TP), using real case studies from national household surveys.
- Provided personalized academic support and continuous student monitoring throughout the semester.
- Supervised final-year projects (PFE) and worker internships, with mentoring on methodological and technical aspects.

Assistant Engineer Intern – Mining Department, Geotechnical Service – Kinross Gold Corporation (Tasiast Mauritania)

📅 July 2025 – September 2025

📍 Tasiast, Mauritania

- **Wall Compliance & 80/80:** designed a Python/Gradio mini-dashboard for *Toe/Crest/Berm Width/BFA* monitoring from CALC and COMPLIANCE files (auto-detection of inputs, KPIs, 80/80 bars, trends & deltas).
- **Daily Survey digitization:** Microsoft *Forms* → *SharePoint/Excel Web* → *Power BI* pipeline with typing/validation, duration calculations, KPIs (% Done, Backlog, SLA Breach), and near real-time refresh.
- **Short-term PV forecasting:** preparation of minute-level data (*lags*, ramps ΔP , time encodings) and sequential prototypes (LSTM/GRU) to anticipate rapid power drops and improve alert *lead time*.
- **Practices:** git versioning, technical documentation, skills transfer to field/process teams.
- **Environment:** Python (NumPy, pandas, Plotly, Gradio), Power BI, SharePoint/Excel Web.

President & Event Coordinator – ESP Data Club

📅 October 2022 – April 2025

📍 Nouakchott, Mauritania

- **Event Coordinator** (October 2022 – January 2023): Organized events to promote statistics and data analysis. Coordinated with members to ensure the success of activities.
- **President** (January 2023 – April 2025): Led the club, set strategic objectives, managed data collection and analysis projects, developed academic partnerships.
- Skills: Leadership, project management, event planning, communication.
- See the Club.

End-of-Study Internship – Caisse des Dépôts et de Développement (CDD)

📅 February 2023 – June 2023

📍 Nouakchott, Mauritania

- Analyzed banking data from CDD to assess the impact of the credit guarantee system on loan defaults in Mauritania.
- Used econometric models and modern machine learning approaches to predict default risks.
- Prepared and interpreted statistical reports on the socio-demographic and financial characteristics of borrowers.
- Developed computational solutions for risk analysis using algorithms such as logistic regression, k-NN, and Random Forest.
- See the analyses.

Enumerator – Agence Nationale de la Statistique et de l'Analyse Démographique et Économique (ANSADE)

📅 July 2022 – August 2022

📍 Nouakchott, Mauritania

- Collected field data for the pilot General Population and Housing Census (RGPH).
- Skills acquired: data collection, communication, fieldwork.

Worker Internship – Agence Nationale de la Statistique et de l'Analyse Démographique et Économique (ANSADE)

📅 April 2022 – June 2022

📍 Nouakchott, Mauritania

- Developed a methodology for data mapping to design sectoral accounts.
- Participated in the collection and processing of economic branch data for national accounts.
- Analyzed sectoral data using ERETES software to compile economic accounts.
- Trained on GDP decomposition techniques as part of the International Comparison Program.
- Contributed to the national survey process for the mobilization and quantification of ZAKAT, including enumerator training.

PROJECTS

Grain Size Analysis of 316L Microstructure – Academic project: Computer-vision grain size analysis

📅 June 2025 – August 2025

📍 Nouakchott, Mauritania

- Developed a complete image-processing pipeline to segment grain boundaries in 316L stainless steel from 500× micrographs.
- Implemented and compared segmentation methods: classical thresholding, gradient filters, the HED model (Caffe/OpenCV), and a convolutional U-Net trained on real and synthetic data.
- Automatic extraction of grain size in physical units (μm , μm^2) via pixel/micron calibration and computation of statistics (size distribution, axes, circularity).
- Created an interactive interface (Gradio) to load a micrograph, segment grains, and export geometric features to CSV.
- See the GitHub repository.

RAG-WebSystem – Academic project: Generative AI system based on Retrieval-Augmented Generation

2025

📅 March 2025 – April 2025

📍 Nouakchott, Mauritania

- Designed a RAG engine (Retrieval + LLM) able to answer queries by drawing on external documents indexed by vectors.
- Implemented a complete pipeline: ingestion, vectorization (FAISS), semantic search, then contextual answer generation with an open-source model.
- Built a REST API (FastAPI) and a reactive web interface (React + Tailwind) to test, trace, and evaluate the answers.
- See the GitHub repository.

DataMine App – Academic project: Corporate industrial project

📅 November 2023 – Present

📍 Nouakchott, Mauritania

- Designed and developed a collaborative platform for managing and tracking tasks on a geostatistical data project.
- Analyzed geostatistical data to optimize resource management and improve decision-making.
- Created an intuitive user interface for managing and visualizing tasks on GitHub.
- See the project [here](#).

Agricultural Data Analysis System (Smart Greenhouse) – Academic project: Entrepreneurship and innovation project 📅 November 2023 – June 2024

📍 Nouakchott, Mauritania

- Designed and developed an intelligent system for analyzing agricultural data, specifically for greenhouses.
- Implemented IoT sensors for real-time data collection and created interactive dashboards for monitoring.
- Applied machine learning techniques to optimize resource management and improve agricultural yields.
- See the project [here](#).

Disaster Tweets NLP Classification – Academic project: Introduction to big data 📅 January 2022 – February 2022

📍 Nouakchott, Mauritania

- Developed a machine learning model to classify tweets related to natural disasters.
- Processed massive data and applied advanced NLP (Natural Language Processing) techniques.
- Used Python libraries to build and evaluate the model.
- See the project [here](#).

Survey Management Application – Academic project: Advanced databases 📅 December 2022 – January 2022

📍 Nouakchott, Mauritania

- Designed and built a survey management application to facilitate data collection and tracking.
- Backend development with the Django framework and front-end creation in HTML, CSS, and JavaScript.
- See the project [here](#).

Survey on the Perception of Influencers – Academic project: Sociological surveys 📅 January 2022 – March 2022

📍 Nouakchott, Mauritania

- Designed and conducted a survey on the perception of influencers in Mauritanian society.
- Developed the questionnaire on KoboCollect to ensure structured data collection.
- Independently collected and analyzed the data, interpreting results to identify social trends.

Grade Management Application – Academic project: Advanced databases and algorithms 📅 December 2022 – January 2022

📍 Nouakchott, Mauritania

- Designed and built a grade management application for students.
- Backend development with the Flask framework and use of HTML, CSS, and JavaScript for the front-end.
- See the project [here](#).

Determinants of Academic Performance in Mauritania – Academic project: Statistical software 📅 May 2021 – June 2021

📍 Nouakchott, Mauritania

- Analyzed the factors influencing academic performance in Mauritania using R.
- Studied data from CONFEMEN's education systems (PASEC1), including socio-demographic, geographic, and pedagogical variables.

SOFT SKILLS

Initiative • Curiosity • Interpersonal Ease • Creative • Listening Skills

EDUCATION

Higher Polytechnic School of Nouakchott – ESP - Mauritania 📅 October 2023 – June 2026

Degree: State Engineer in Statistics and Data Engineering.

National School of Computer Science and Systems Analysis – ENSIAS - Rabat 📅 October 2025 – February 2026

Exchange semester: Computer Engineering and Networks.

Faculty of Economic Sciences and Management – FSEG - Mauritania 📅 October 2023 – June 2025

Degree: Master's in Applied Econometrics and Statistics – Economic Statistical Engineering Track.

Polytechnic Group – Higher Institute of Statistical Professions – GP - Mauritania 📅 October 2020 – June 2023

Degree: Professional Bachelor's in Applied Statistics.

Atar High School – Atar – Mauritania

2019 – 2020

Degree: Baccalaureate, Mathematics track (Bac C).

SKILLS

- Analysis & Modeling:** statistics, econometrics, surveys; data mining (classification, regression, clustering, association rules); time series (short-term forecasting).
- Big Data, Data Engineering, DB & GIS:** Architecture & modeling of a centralized Data Warehouse, ELT/ETL pipelines (Airbyte, dbt), data flow orchestration (N8N), Hadoop, Spark, Kafka, MapReduce, Hive SQL; SQL management & optimization (execution plans, indexing, partitioning, Oracle SQL Tuning); Docker & Dev Containers; Stata, SPSS, Excel, GAMS, Eviews, Sphinx, KoboToolbox, ODKCollect, CSPro; Oracle SQL Developer; PostgreSQL/PostGIS (incl. pgvector); QGIS/ArcGIS; GIS integration & analysis (spatial joins, basic geoprocessing).
- AI, MLOps & Programming:** supervised ML (regression, SVM, random forests, decision trees); deep learning (LSTM/GRU for time series, CNN, Transformers) with TensorFlow, PyTorch, Keras; NLP & RAG (embeddings, language models, augmented retrieval, RAGAI API); MLOps (MLflow, Docker/Compose); Python/PySpark, Julia, R, Scala, SQL (incl. HiveQL), PL/SQL, Java (J2EE, **Spring Boot, Spring Cloud, Spring Data, Spring Security, Spring microservices**), VBA, $\text{\LaTeX}$; Bash & Linux CLI; Git/GitHub; Jira, Trello, VS Code, Cursor; web development (HTML, CSS, JavaScript, Django, Flask, REST APIs, J2EE middleware integration), React, Flutter.

LANGUAGES

Arabic Native Language	
French DALF C1	
English CEFR B1 - Tracktest English Assessment	

VOLUNTEERING

- President, ISS Alumni Association (current position)
- Events & Programs Officer, Association des Statisticiens Mauritaniens
- President, ESP Data Club (January 2023 – April 2025)
- Event Coordinator, ESP Data Club (October 2022 – January 2023)
- Member, Clubs Leaders Étudiants Francophones (CLEF)
- Member, Mauritanian Engineering Community (MEC)
- Military Training

ACHIEVEMENTS

-  **1st Place, RIMAI Data Science Challenge 2024**
Competition focused on license plate recognition using AI techniques. Learn more about the challenge at the RIMAI Challenge Website.
-  **Organizer, First Datathon in Mauritania**
Three-day competition centered on Mauritanian data analysis. As president of the ESP Data Club, I supervised the organization and judged competitors through competitive data science challenges.

CERTIFICATIONS

freeCodeCamp (6): Relational Database (v8) (2025) • Data Visualization (2023) • Data Analysis with Python (2022) • Scientific Computing with Python (2022) • Front End Development Libraries (2022) • Responsive Web Design (2022)

DataCamp (9): Data Manipulation in SQL (2023) • Intermediate SQL (2023) • Introduction to Data Science in Python (2023) • Python Data Science Toolbox (Part 2) (2023) • Understanding Data Engineering (2023) • Writing Efficient R Code (2023) • Intermediate Python (2022) • Intermediate R (2022) • Introduction to Version Control with Git (2022)

Google (3): Google Advanced Data Analytics Capstone (2023) • Google Advanced Data Analytics Professional Certificate (2023) • Google Data Analytics (2023)

INTERESTS

- Coding
- Travel
- Football
- Tourism